
Bayesian Filter

§ 1.1

Basic Settings

Let

$$\mathbf{x}_t \in \mathbb{R}^{n_x}$$

denote the state vector at time t , which represents the unknown quantity we want to infer.

At each time t , we observe

$$\mathbf{z}_t \in \mathbb{R}^{n_z}.$$

The observation contains information about the hidden state $\mathbf{x}_t$, but may be noisy and uncertain. For convenience, define the observation history up to time t as

$$\mathbf{z}_{1:t} = \{\mathbf{z}_1, \dots, \mathbf{z}_t\}$$

Similarly, suppose that a control input is available at each time t , denoted by

$$\mathbf{u}_t \in \mathbb{R}^{n_u},$$

with the corresponding control history

$$\mathbf{u}_{1:t} = \{\mathbf{u}_1, \dots, \mathbf{u}_t\}.$$

Therefore, the available information evolves sequentially as

$$\mathbf{u}_1, \mathbf{z}_1, \mathbf{u}_2, \mathbf{z}_2, \mathbf{u}_3, \mathbf{z}_3, \dots$$

whereas the states

$$\mathbf{x}_1, \mathbf{x}_2, \dots$$

are not directly observed.

Since subsequent inference depends on the state, the objective is to characterize the uncertainty associated

with the random variable $\mathbf{x}_t$ given all available information. This uncertainty is represented by the belief state

$$\text{bel}(\mathbf{x}_t) = p(\mathbf{x}_t | \mathbf{z}_{1:t}, \mathbf{u}_{1:t}).$$

The belief state is therefore a probability distribution over the possible values of $\mathbf{x}_t$ conditioned on the available observation and control histories. It is also commonly referred to as the posterior state distribution, information state, or state of knowledge.

§ 1.2

Bayesian Filter Derivation

The key idea of Bayesian filter is to represent uncertainty explicitly using the calculus of probability theory.

At each time step, the hidden state $\mathbf{x}_t$ evolves from the previous state $\mathbf{x}_{t-1}$ under the control input $\mathbf{u}_t$, while the observation $\mathbf{z}_t$ is generated from the corresponding state $\mathbf{x}_t$, as shown in Figure 1.1.

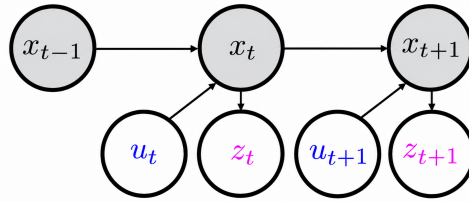


Figure 1.1 Graphical representation of the Bayesian filtering model

However, the belief, $p(\text{current state} | \text{all past information})$, is intractable. This leads to the Markov assumption: future state conditionally independent of past control inputs and observations given present state. We therefore have

$$p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) = p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_t),$$

$$p(\mathbf{z}_t | \mathbf{x}_t, \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) = p(\mathbf{z}_t | \mathbf{x}_t).$$



Using Bayes' rule and the Markov assumptions,

$$\begin{aligned}
\text{bel}(\mathbf{x}_t) &= p(\mathbf{x}_t | \mathbf{z}_{1:t}, \mathbf{u}_{1:t}) = p(\mathbf{x}_t | \mathbf{z}_t, \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) \\
&= \frac{p(\mathbf{x}_t, \mathbf{z}_t | \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1})}{p(\mathbf{z}_t | \mathbf{u}_t)} \leftarrow \text{Condition probability} \\
&= \eta p(\mathbf{x}_t, \mathbf{z}_t | \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) \leftarrow \text{Normalization constant} \\
&= \eta p(\mathbf{z}_t | \mathbf{x}_t, \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) p(\mathbf{x}_t | \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) \leftarrow \text{Bayes rule} \\
&= \eta p(\mathbf{z}_t | \mathbf{x}_t) p(\mathbf{x}_t | \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) \leftarrow \text{Markov assumption} \\
&= \eta p(\mathbf{z}_t | \mathbf{x}_t) \int p(\mathbf{x}_t, \mathbf{x}_{t-1} | \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) dx_{t-1} \leftarrow \text{Opposite of marginalization} \\
&= \eta p(\mathbf{z}_t | \mathbf{x}_t) \int p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) p(\mathbf{x}_{t-1} | \mathbf{u}_t, \mathbf{z}_{1:t-1}, \mathbf{u}_{1:t-1}) dx_{t-1} \leftarrow \text{Probability chain rule} \\
&= \underbrace{\eta p(\mathbf{z}_t | \mathbf{x}_t)}_{\text{correction}} \underbrace{\int p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_t) \text{bel}(\mathbf{x}_{t-1}) dx_{t-1}}_{\text{prediction}}
\end{aligned}$$

where η is a normalization constant and $\eta = \frac{1}{p(\mathbf{z}_t | \mathbf{u}_t)}$ and $\text{bel}(\mathbf{x}_0) = p(\mathbf{x}_0)$.

Bayesian filter therefore provides a recursive update approach to calculate belief tractably consisting of two steps: prediction, which propagates the previous belief through the motion model, and correction, which incorporates the latest observation through the measurement model.